# Applications of differentiation
# MTH 201 - Business Calculus

**Phan Xuân Thành**

Faculty of Mathematics and Informatics
Hanoi University of Science and Technology
e-mail: thanh.phanxuan@hust.edu.vn




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# Outline


1. **Using first derivatives to find maximum and minimum values**
2. Using second derivatives to find maximum and minimum values and test for concavity
3. Graph sketching: Asymptotes and rational functions
4. Using derivatives to find absolute maximum and minimum values

# Increasing/Decreasing functions

If the graph of a function rises from left to right over an interval $I$, the function is said to be increasing on, or over, $I$.



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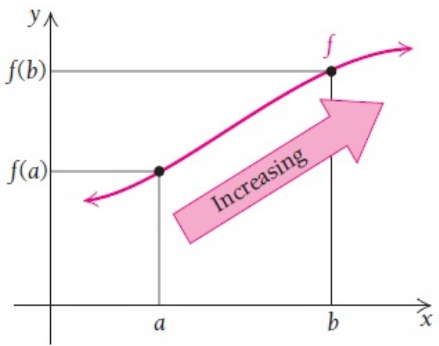
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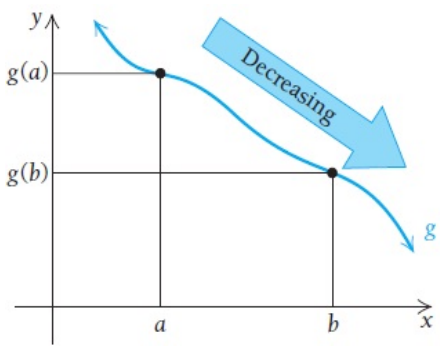
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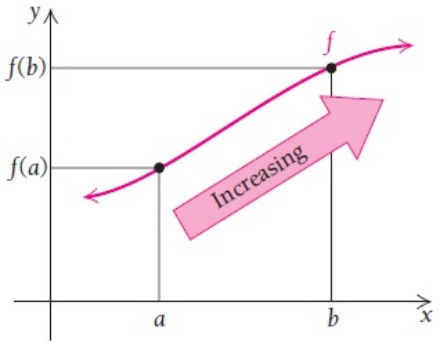


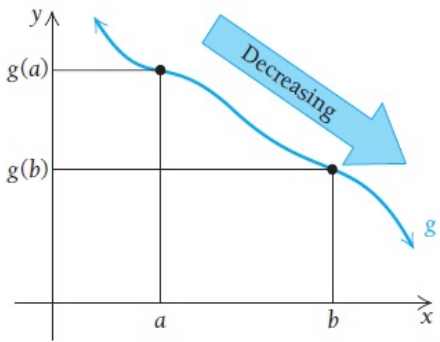

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A function $g$ is decreasing over $I$ if, for every $a, b \in I$, and $a < b$, then $g(a) > g(b)$.

# Increasing/Decreasing functions

**Theorem (Increasing/Decreasing Test)**

*Let $f$ be differentiable over an open interval $I$.*
*If $f'(x) > 0$ on $I$, then $f$ is increasing on that interval.*
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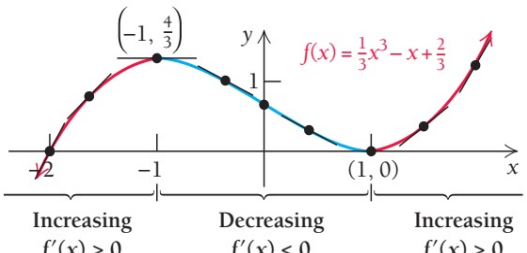
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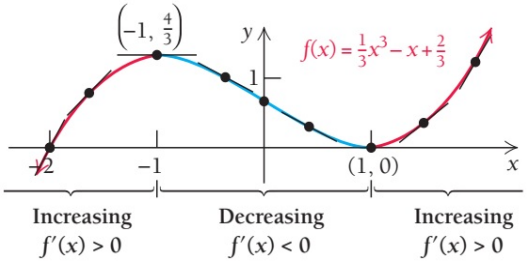

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Example. Let $f(x) = \frac{1}{3}x^3 - x + \frac{2}{3}$.




Exercise. For what values of $c$ is the function $f(x) = \ln(x^2 + 1) - cx$ increasing on $(-\infty, \infty)$?

# Relative maximum and minimum values

**Definition**

*Let $D$ be the domain of $f$. $f(c)$ is a relative minimum if there exists within $D$ an open interval $I$ containing $c$ such that $f(c) \leq f(x)$, for all $x$ in $I$;*
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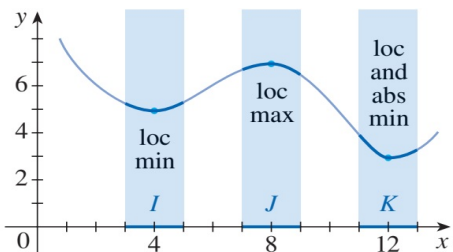
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# Critical values

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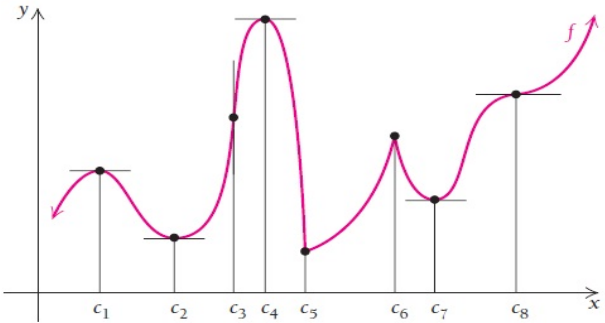
*A critical value of a function $f$ is any number $c$ in the domain of $f$ for which the tangent line at $(c, f(c))$ is horizontal or for which the derivative does not exist.*
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Consider the graph of a continuous function $f$ in the following figure






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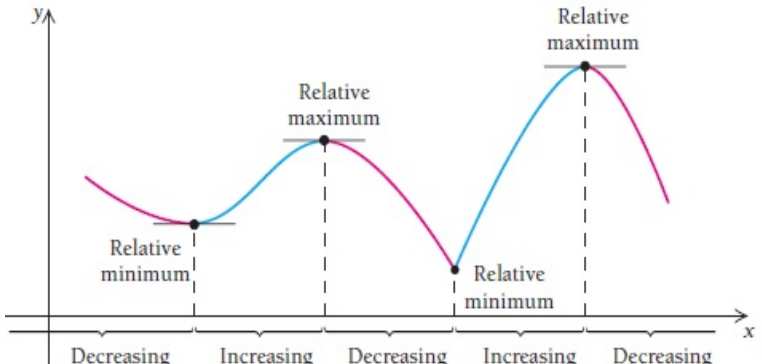
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**Theorem (First derivative test for relative extrema)**

*Suppose that $c$ is a critical value of a continuous function $f$, and that $f$ is differentiable at every point in some interval containing $c$ except possibly at $c$ itself. Moving across $c$ from left to right,*

1. *if $f'$ changes from negative to positive at $c$, then $f$ has a local minimum at $c$. That is, $f$ is decreasing to the left of $c$ and increasing to the right of $c$;*
2. *if $f'$ changes from positive to negative at $c$, then $f$ has a local maximum at $c$. That is, $f$ is increasing to the left of $c$ and decreasing to the right of $c$;*
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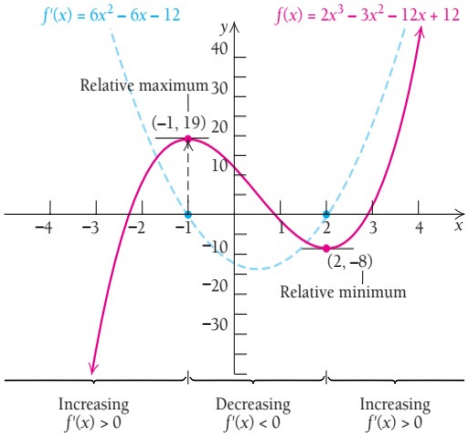
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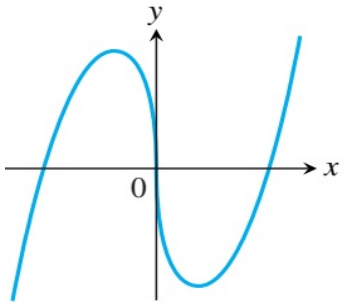
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$f$ does not have an extreme value at $x = 0$; $f$ has a local minimum at $x = 1$, the value of the local minimum is $f(1) = -6$; $f$ has a local maximum at $x = -1$, $f(-1) = 6$.



# Relative maximum and minimum values

Exercise 2. Find the relative extrema of $y = x^{1/3}(x-4)$.



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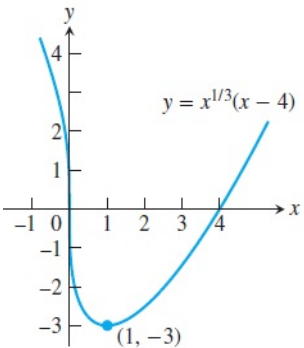
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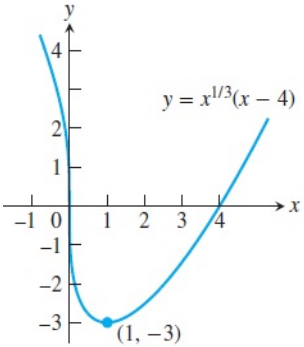





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Exercise 3. Find any relative extrema of each function

a) $f(x) = 3x^4 - 15x^2 + 2$ b) $g(x) = \dfrac{3x}{x^2+4}$ c) $h(x) = x\sqrt{8-x^2}$ d) $p(x) = 4\ln x - x^3$.

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# Classifying relative extrema using second derivatives

## Second derivative test for relative extrema

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Second derivative test for relative extrema

**Theorem**

*Suppose $f''$ is continuous on an open interval that contains $x = c$ and $f'(c) = 0$.*

1. *If $f''(c) < 0$, then $f$ has a relative maximum at $x = c$.*
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Example 1. Find any relative extrema of each function. State where the function is increasing or decreasing.

a) $f(x) = 20x^3 - 3x^5$ b) $g(x) = 4x - 6x^{2/3}$ c) $h(x) = (x-1)^{2/3} - (x+1)^{2/3}$.

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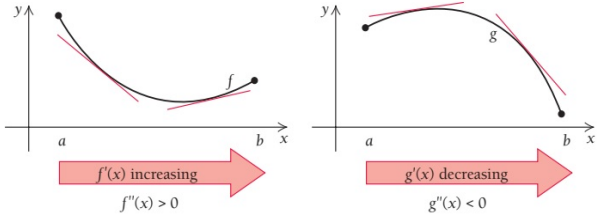
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Example 2. Find the relative extrema of the function $f$ given by $f(x) = \dfrac{\ln x}{x}$, then show that $\pi^e < e^\pi$.

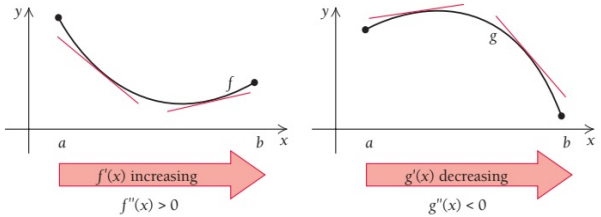
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## Increasing and decreasing derivatives



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*The graph of a differentiable function $y = f(x)$ is*

**i)** *concave up on an open interval $I$ if $f'$ is increasing on $I$.*

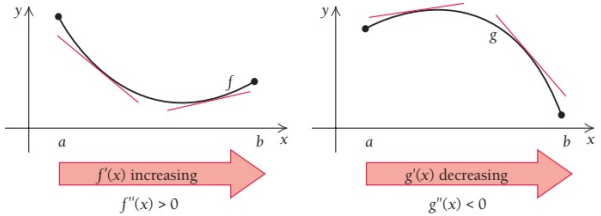
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If the graph of $y = f(x)$ is concave up on $I$ then all the tangent lines for $f$ are below the graph.

# Concavity

**Theorem (Test for concavity)**

1. *If $f''(x) > 0$ on an interval $I$, then the graph of $f$ is concave up. ($f'$ is increasing, so $f$ is turning up on $I$.)*
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Example. Determine the concavity of

a) $f(x) = 80 - 9x^2 - x^3$ b) $g(x) = \dfrac{x}{x^2+1}$ c) $h(x) = x^4 - 4x^3 + 10$.

# Points of inflection

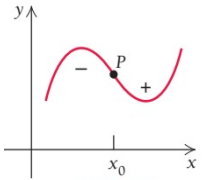
**Definition (Point of inflection)**

*A point where the graph of a function has a tangent line and where the concavity changes is a point of inflection.*

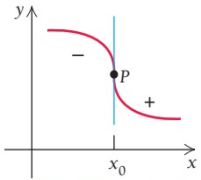
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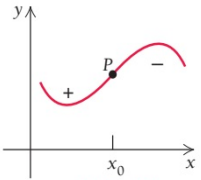
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$f''(x_0) = 0$




$f''(x_0)$ does not exist.

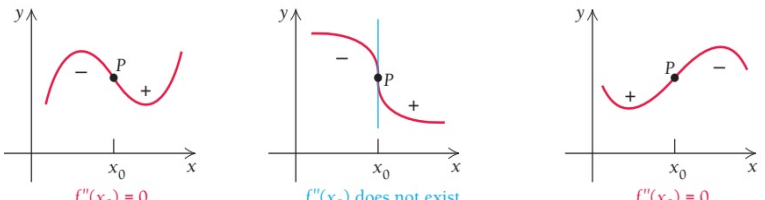



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# Points of inflection

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## Theorem (Finding points of inflection)

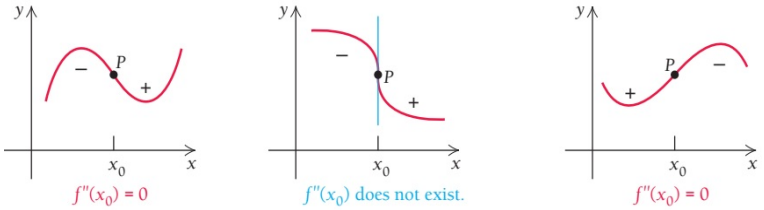
*If a function $f$ has a point of inflection, it must occur at a point $x_0$, in the domain of $f$, where*

$$f''(x_0) = 0 \quad \textit{or} \quad f''(x_0) \ \textit{does not exist.}$$



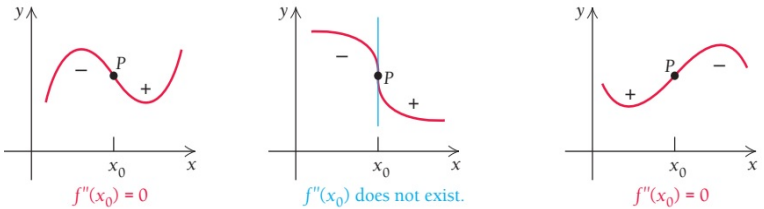
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# Points of inflection




To find candidates for points of inflection, we look for numbers $x_0$ for which $f''(x_0) = 0$ or for which $f''(x_0)$ does not exist. Then, if $f''(x)$ changes sign as $x$ moves through $x_0$, we have a point of inflection at $x_0$.

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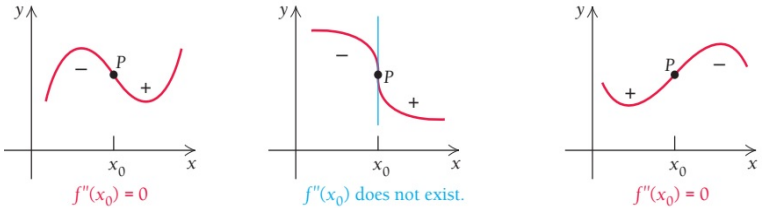

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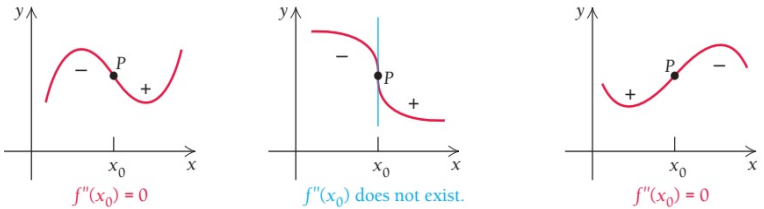
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The curve $y = x^{1/3}$ has a point of inflection at $x = 0$, but $y''$ does not exist there.



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# Points of inflection

Example 2. Use the second derivative to determine the point(s) of inflection for the function $f(x) = 3x^5 - 20x^3$.

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Answer.



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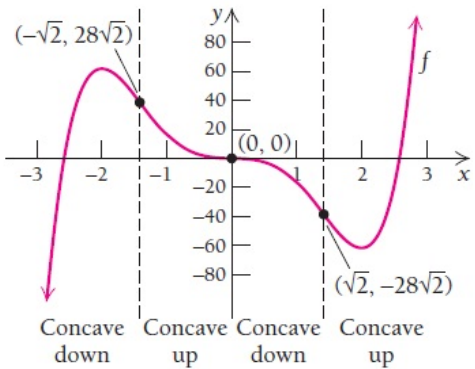
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Answer. We have $f'(x) = 15x^4 - 60x^2$ and $f''(x) = 60x^3 - 120x$

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# Exercises

Exercise. For each function, list the coordinates of any extrema or points of inflection. State where the function is increasing or decreasing, where its graph is concave up or concave down

a) $f(x) = 8x^3 - 6x + 1$ b) $f(x) = x^4 - 6x^2$

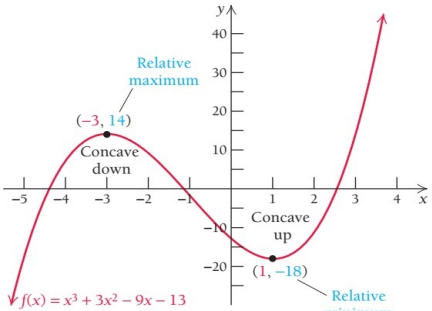
c) $f(x) = x\sqrt{4 - x^2}$ d) $f(x) = (x + 1)e^{-x}$ e) $f(x) = x^3 + 3x^2 - 9x - 13.$

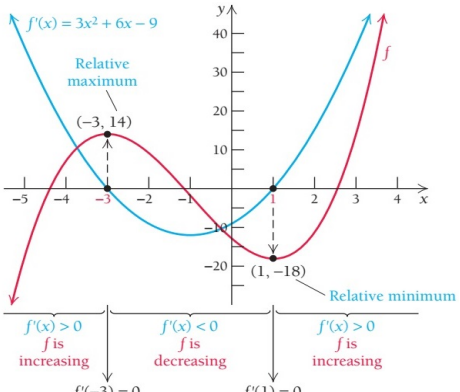
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# Outline


1. Using first derivatives to find maximum and minimum values
2. Using second derivatives to find maximum and minimum values and test for concavity
3. **Graph sketching: Asymptotes and rational functions**
4. Using derivatives to find absolute maximum and minimum values

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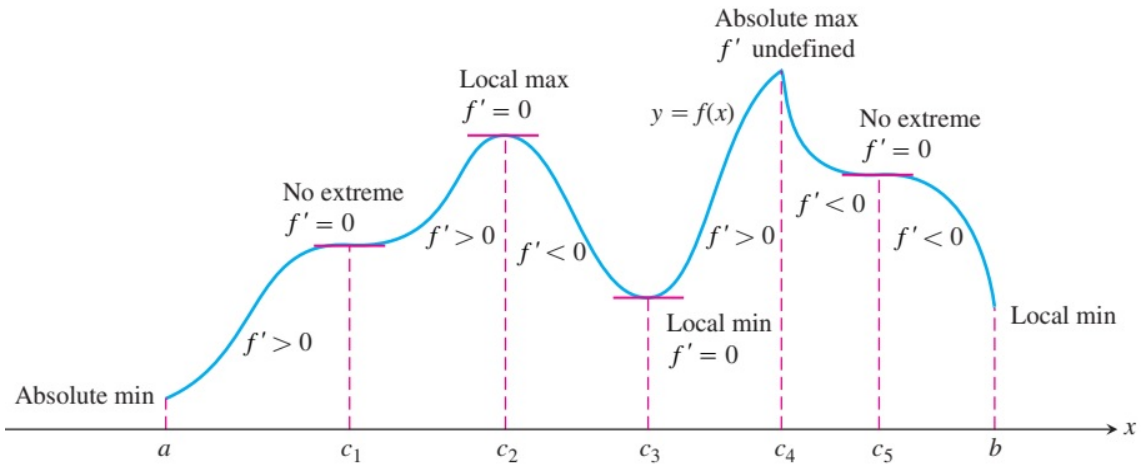

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Problem. *How to find absolute maximum and minimum values over closed intervals?*

**Theorem**

*If $f$ is continuous on a closed interval $[a, b]$, then $f$ attains both an absolute maximum value $M$ and an absolute minimum value $m$ in $[a, b]$. That is, there are numbers $x_1$ and $x_2$ in $[a, b]$ with $f(x_1) = m$, $f(x_2) = M$ and $m \leq f(x) \leq M$ for every other $x$ in $[a, b]$.*



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# Finding absolute maximum and minimum values

**Theorem**

*Suppose $f$ is a continuous function defined over a closed interval $[a, b]$. To find the absolute maximum and minimum values over $[a, b]$:*
*a) First find $f'(x)$.*
*b) Then determine all critical values in $[a, b]$. That is, find all $c$ in $[a, b]$ for which $f'(c) = 0$ or $f'(c)$ does not exist.*
*c) List the values from step (b) and the endpoints of the interval: $a, c_1, c_2, ..., c_n, b$.*
*d) Evaluate $f(x)$ for each value in step (c): $f(a), f(c_1), f(c_2), ..., f(c_n), f(b)$.*
*The largest of these is the absolute maximum of $f$ over $[a, b]$. The smallest of these is the absolute minimum of $f$ over $[a, b]$.*

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Example. Find the absolute maximum and minimum values of $f(x) = x^4 - 2x^3$ over the interval $[-2, 2]$.



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**Theorem**

*Suppose $f$ is a function such that $f'(x)$ exists for every $x$ in an interval $I$ and there is exactly one (critical) value $c$ in $I$, for which $f'(c) = 0$. Then*

$$f(c) \text{ is the absolute maximum value over } I \text{ if } f''(c) < 0$$

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Example. Find the absolute maximum and minimum values of each function over the indicated interval

a) $f(x) = x^2 + \dfrac{432}{x}$ over $(0, \infty)$

b) $g(x) = -x^3 + x^2 + 5x - 1$ over $(0, \infty)$

c) $h(x) = \frac{1}{3}x^3 + 2x^2 + x$ over $[-4, 0]$.



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# Finding absolute maximum and minimum values

Exercise 1. For a dosage of $x$ cubic centimeters (cc) of a certain drug, the resulting blood pressure $B$ is approximated by $B(x) = 305x^2 - 1830x^3$, $0 \leq x \leq 0.16$. Find the maximum blood pressure and the dosage at which it occurs.

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Exercise 2. The percentage of unemployed workers in service occupations can be modeled by $p(x) = -0.039x^3 + 0.594x^2 - 1.967x + 7.555$, where $x$ is the number of years since 2003. (*Source*: Based on data from www.bls.gov.) According to this model, in what year during the period 2003–2013 was this percentage a maximum?



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Exercise 3. The total-cost and total-revenue functions for producing $x$ items are $C(x) = 5000 + 600x$ and $R(x) = -\frac{1}{2}x^2 + 1000x$, where $0 \leq x \leq 600$.
a) Find the total-profit function $P(x)$.
b) Find the number of items, $x$, for which total profit is a maximum.
c) Average profit is given by $A(x) = P(x)/x$. Find $A(x)$.
d) Find the number of items, $x$, for which average profit is a maximum.